

Probability and applications

Student investigations / Remaining atlas entries / September 2026

Choose an investigation whose tools you know. Make things, try ideas and explain what you notice. Use scrap paper or work with a partner. Later pages can be another meeting.

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Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

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Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Which messenger should change your mind?

The same clue, a different experiment

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Bag A holds R,R,B; bag B holds R,B,B. For each experiment choose one bag by a fresh fair coin. Messenger DRAWS takes one uniformly random counter. Messenger CHOOSES looks inside and deliberately shows a red counter whenever one is present. You hear “red” from one specified messenger.

1. Choose a messenger. Before calculating, predict whether red favors A, favors B, or leaves them equally likely. Build the possible bag-and-counter outcomes for DRAWS. Why may you give those cards equal weight?

Bag A

A1 · R	A2 · R	A3 · B
--------	--------	--------

Equal-weight outcomes that survive

Bag B

B1 · R	B2 · B	B3 · B
--------	--------	--------

Equal-weight outcomes that survive

2. Find the chance of A after each messenger says red. Explain why the identical word did not force an identical answer.

Your construction / explanation

Ask the same bag twice

The same clue, a different experiment

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Choose A or B once by a fair coin. Draw uniformly, record the color, REPLACE the counter, mix, and draw again independently from that same bag. Each bag has nine labeled ordered draw pairs. The bag is not reselected between draws.

3. Use the two 3×3 grids to locate every way to hear RR. What chance of A does that clue give? Compare it with a single red draw.

	Bag A		
	A1	A2	A3
A1			
A2			
A3			

	Bag B		
	B1	B2	B3
B1			
B2			
B3			

Rows: first counter Columns: second counter

4. Choose one new report: “red then blue,” “the two colors differ,” or “at least one red.” Find its answer, then trade proofs with a group that chose another report.

Your construction / explanation

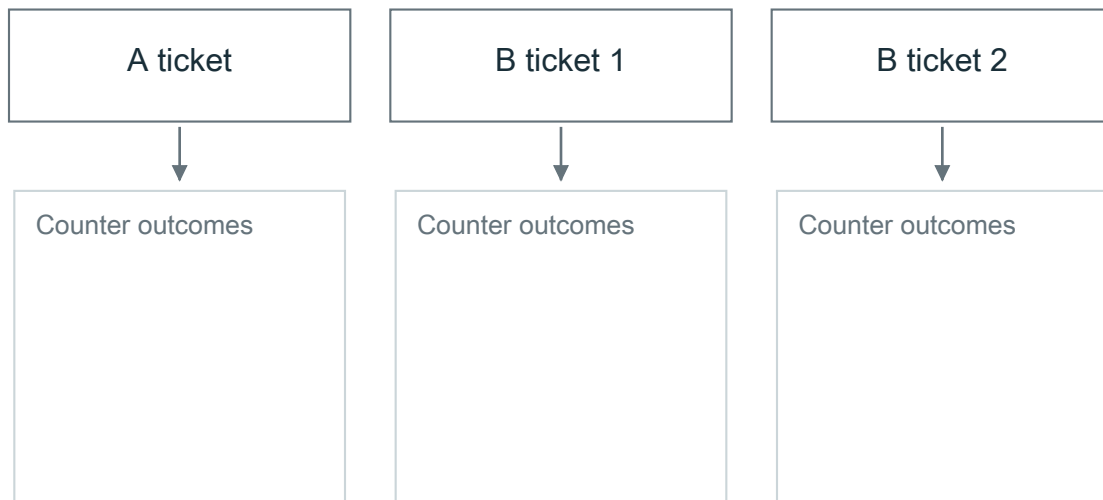
Can you cancel a clue?

The same clue, a different experiment

Count equally likely outcomes; fractions; distinguish drawing from reporting.

Now select the bag by shuffling three equally likely tickets A,B,B and taking one. Then make ONE uniform counter draw. Every ticket-and-counter pair is equally likely; the two B tickets remain distinct.

5. After a red draw, are A and B equally likely? Build a card proof before using a formula.



6. Invent two equally sized bags that make a random red draw leave a fair bag choice unchanged. Make another pair where red favors A as strongly as possible while BOTH bags can produce red. State the bag size you chose.

Red changes nothing

A

B

Common size: _____

Red favors A most

A

B

Common size: _____

Assign the clips before explaining the result

How surprising are these labels?

Small subset counts and means; explicit null model and random assignment.

Six fictional pilots have scores $A=0, B=1, C=2, D=4, E=5, F=6$. Before flying, exactly three received TRAINED labels, uniformly among all three-person subsets. The observed trained group is D,E,F. For this investigation ASSUME training changed no pilot's score. Keep scores fixed and move only labels.

1. Try three possible assignments. Use the sum of trained scores to say which looks strongest. Why can sums replace means here?

A = 0

B = 1

C = 2

D = 4

E = 5

F = 6

Your three trained groups / sums / comparison

2. Find ALL possible assignments without listing any twice. Try putting A in one group, then pairing that assignment with the assignment that swaps the two group labels. Explain why this is complete.

Paired assignments / sum on each side - organize the list your way

Choose the alarm rule before looking

How surprising are these labels?

Small subset counts and means; explicit null model and random assignment.

Before data, choose the statistic and direction. UP means larger trained sums are more extreme. EITHER WAY means larger absolute trained-minus-other mean differences are more extreme. Now the observed trained sum is 15 and the mean difference is 4; compare all assignments against these observed values.

3. Under the no-effect assumption, what fraction of assignments trigger UP? What fraction trigger EITHER WAY? Could an UP tail ever have probability smaller than $1/20$ in this exact design?

Your construction / explanation

4. A reporter waits to see which group did better, then advertises the one-sided $1/20$ count in that direction. Under no effect, how often would this “pick the favorable direction” procedure flag either extreme? Explain what the reporter left out.

Your construction / explanation

5. What have the counts established-and what have they NOT established? Include the assumption that made scores stay fixed.

Your construction / explanation

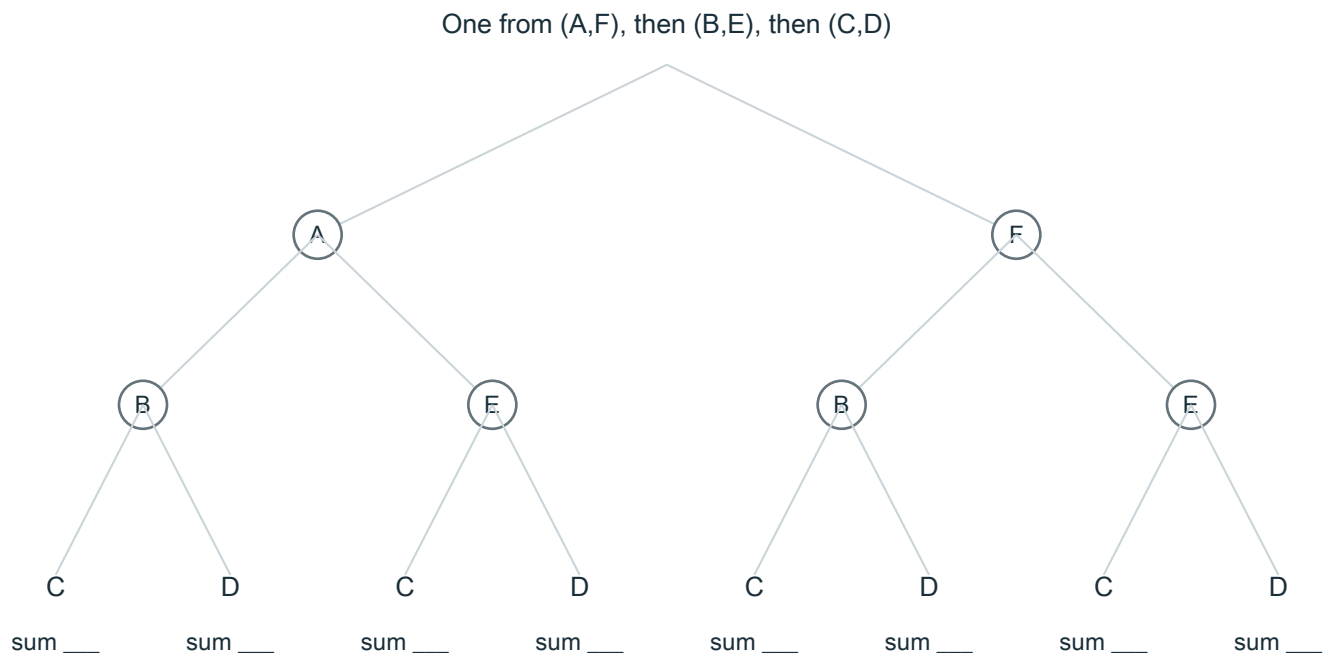
A new assignment rule changes the answer

How surprising are these labels?

Small subset counts and means; explicit null model and random assignment.

Instead, form fixed pairs (A,F),(B,E),(C,D). Before the flights, use an independent fair coin in EACH pair to select exactly one trained pilot. Keep the same sharp no-effect assumption and observed trained group D,E,F.

6. List the possible trained groups using a three-coin tree. How often is their total at least 15 now? Why would using the old twenty-outcome map be wrong?



7. Design a one-sentence explanation for someone who says, “The scores stayed the same, so the probability had to stay the same.”

Your construction / explanation

Could two different worlds look identical?

A very precise wrong answer

Small population means; distinguish the sampling frame from the target.

The target is the mean of ALL eight cards. Cards 1-4 are 1,1,1,1. Cards 5-8 are hidden. The sampler chooses an identifier uniformly only from 1-4, reads its value, replaces it, and repeats independently. You may request any number of these readings.

1. Make two different possible sets of hidden cards 5-8 that produce exactly the same readings from this sampler but have different whole-population means. Use values from 0 to 9.

	1	2	3	4	5	6	7	8
Known	1	1	1	1	?	?	?	?
World 1	1	1	1	1				
World 2	1	1	1	1				

Whole means: World 1 _____ World 2 _____

2. Could a million readings from this same sampler distinguish your two worlds? Propose one change to what may be observed that could help. State exactly what it would establish.

Your construction / explanation

A noisy rule with the right center

A very precise wrong answer

Small population means; distinguish the sampling frame from the target.

Now use this supplied population: values 1,1,1,1,5,5,5,5. New rule FULL draws an identifier uniformly from 1-8, replaces it, and draws again independently. Its estimate is the mean of the two values.

3. Try to predict all possible two-draw means. Find their exact probabilities without writing 64 individual ordered identifier pairs. Is the expected estimate the whole-population mean?

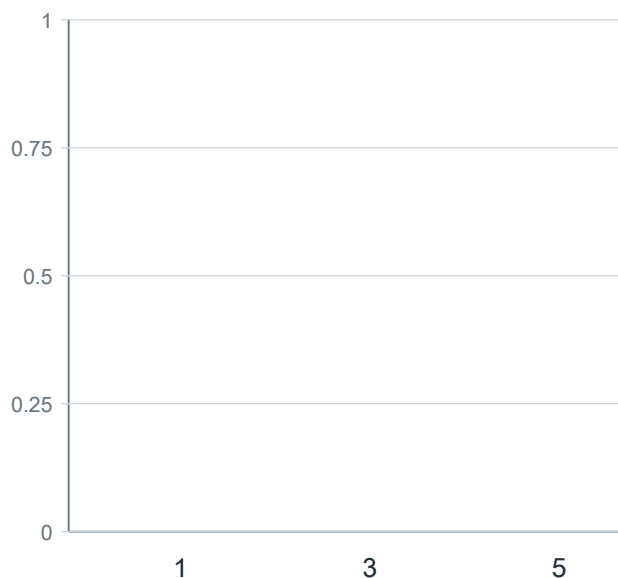
Group equal-value outcomes instead of listing all identifiers.

Draw 1 / draw 2

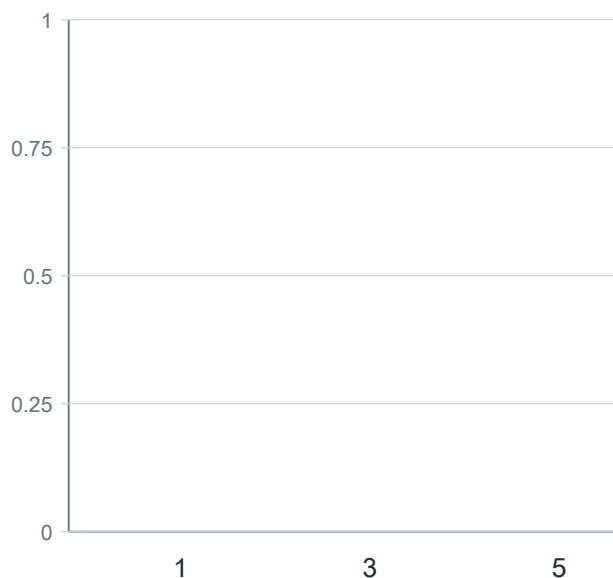
Means / weights / expected value

4. A FULL sample happens to give 1, just like the restricted sampler. Did that one result prove the two procedures are equally trustworthy? Draw their complete distributions to explain.

FULL



RESTRICTED



Design a better two-card survey

A very precise wrong answer

Small population means; distinguish the sampling frame from the target.

Continue with the supplied population of four 1-cards and four 5-cards. Divide the identifiers into blocks 1-4 and 5-8. A BLOCK sampler draws one uniformly within each block, independently, then averages the two readings.

5. Compare BLOCK with FULL. What does BLOCK estimate on every possible draw? Explain why this certainty depends on this population rather than on a magic property of sampling one per block.

Block L

1	1	1	1
---	---	---	---

One draw / its contribution

Block R

5	5	5	5
---	---	---	---

One draw / its contribution

6. Change the population to TWO 1-cards in block L and SIX 5-cards in block R. A sampler still takes one card per block. Is their ordinary average the whole mean? Repair the rule using the known block sizes.

Block L

1	1
---	---

One draw / its contribution

Block R

5	5	5	5	5	5
---	---	---	---	---	---

One draw / its contribution

What does the strip catch?

Build a promise an interval can keep

Integer intervals and fractions; track one fixed secret across random errors.

A secret integer θ stays fixed. The instrument reports $X=\theta+E$, where E is freshly chosen uniformly from $-2,-1,0,1,2$. Try the rule $[X-1,X+1]$, including both endpoints. The interval moves with X ; the secret does not move.

1. Choose a practice secret and run all five possible errors once, as an exact audit rather than a random sample. Place each reported interval. Which catch your secret?

My practice secret $\theta = \underline{\hspace{2cm}}$ Label each scale before placing its strip.

E = -2		catch? ____
E = -1		catch? ____
E = 0		catch? ____
E = 1		catch? ____
E = 2		catch? ____

2. Without redoing five calculations, explain why moving the secret to another integer leaves the success fraction unchanged. What assumption about the errors did your explanation use?

Your construction / explanation

The shortest four-out-of-five promise

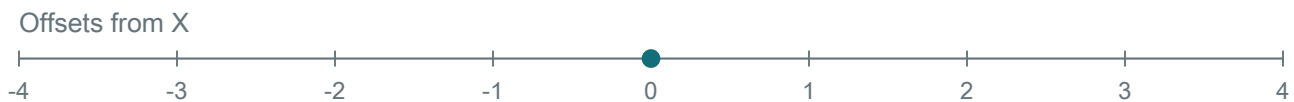
Build a promise an interval can keep

Integer intervals and fractions; track one fixed secret across random errors.

Restrict to translation rules $[X+a, X+b]$, where $a \leq b$ are fixed integers chosen BEFORE seeing X . Width means $b-a$, not the number of integer dots. The same offsets must be used for every reading.

3. Design a rule with coverage at least $4/5$ for EVERY integer secret. Make its width as small as possible. You may put the reading off-center. Find every shortest rule.

Compare candidate intervals above or below the same offset scale



4. Prove that a narrower rule cannot meet the promise. Then design a 100% rule and explain exactly which model assumption makes its guarantee possible.

Your construction / explanation

An honest percentage can hide a bad rule

Build a promise an interval can keep

Optional: independent randomization and distinction between coverage and posterior belief.

Optional: after measuring X , flip a fresh independent fair coin. On HEADS report $[X-2, X+2]$. On TAILS report the one-point interval $[X+10, X+10]$. Always reveal which branch you used. Keep the same bounded-error model.

5. What is this procedure's coverage for each fixed secret? Prove your answer using all ten error-and-coin cases.

	HEADS	TAILS
$E = -2$		
$E = -1$		
$E = 0$		
$E = 1$		
$E = 2$		

6. You see $X=7$ and the TAILS report $[17, 17]$. Is it reasonable to call 17 a 50% plausible secret just because the procedure has 50% coverage? Explain what the percentage actually certifies.

Your construction / explanation

7. Would you recommend this 50% rule over your shortest 80% rule? Give a reason beyond simply choosing the larger percentage.

Your construction / explanation

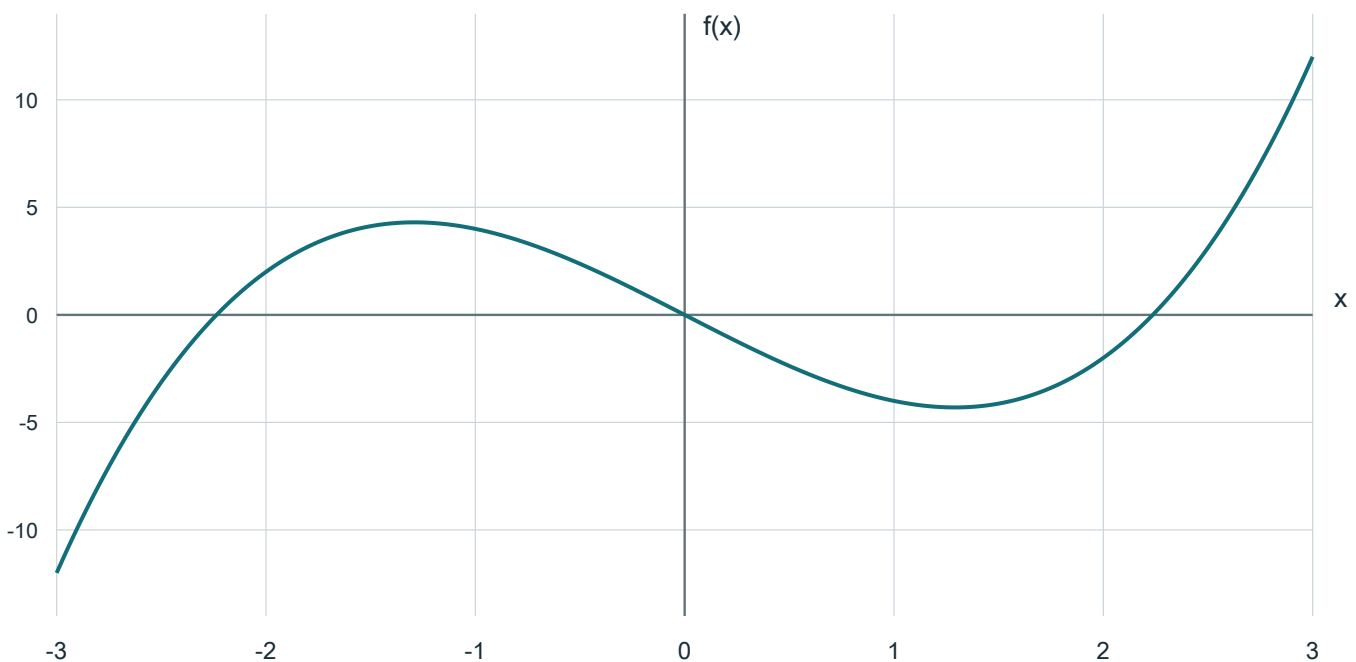
Follow a tangent, not the curve

When the tangent robot comes back

Calculus: derivatives and tangent updates; continuity for the safeguard.

Use $f(x)=x^3-5x$ and $f'(x)=3x^2-5$. From guess x , draw the tangent at $(x, f(x))$; its horizontal-axis crossing is the next guess. Then move vertically to the curve and repeat. The initial guess is x_0 .

1. Choose two starts from $-2, -1, 0, 1, 2$. Draw or calculate two steps for each. Predict what a third step might do. Keep your exact values when they are simple.



Curve supplied; add your own tangents and exact values.

2. Derive a formula for the next guess. For a start at 1, predict x_{1000} and x_{1001} without making a long table. Prove the prediction.



Next-guess formula / proof of later steps

A failed run is not a failed equation

When the tangent robot comes back

Calculus: derivatives and tangent updates; continuity for the safeguard.

Keep the same cubic. A solver must distinguish a true zero, a repeating guess and an undefined tangent step.

3. Find every real root. Find the starts where the Newton formula is undefined. Explain why “the robot failed” does not imply “there is no root.”

Your construction / explanation

4. Design a warning that catches the exact two-cycle. Could a warning alone force the solver to reach a root? Give a separate certificate that a root lies in $[1,3]$.

Your construction / explanation

Give the tangent robot a guardrail

When the tangent robot comes back

Calculus plus continuity and interval inequalities.

Keep a bracket $[L, R]$ with opposite-sign endpoint values and a current guess x . Propose its Newton step. ACCEPT the proposal only if it is defined and lies in the middle half $[L+(R-L)/4, R-(R-L)/4]$. Otherwise use the midpoint. If the chosen point is a root, stop; otherwise retain the subinterval whose endpoints have opposite signs. Use the chosen point as the next current guess.

5. Start with $[L, R] = [1, 3]$ and $x = 1$. Carry out two guarded updates. Record which proposal was accepted and the new bracket each time.

Update 1: bracket _____ current x _____ Newton proposal _____

Accepted? _____ chosen point _____ sign _____ new bracket _____

Update 2: bracket _____ current x _____ Newton proposal _____

Accepted? _____ chosen point _____ sign _____ new bracket _____

6. Prove that the bracket always retains a root and its width is at most $3/4$ of the old width after an update. Give a bound after n updates starting from width 2. How many updates suffice to make the width less than 0.01?

Your construction / explanation

Build it. Try to break it.

How many rooms must a machine remember?

Follow labeled arrows; remainders; compare histories using common futures.

A machine starts in its marked room, reads one symbol at a time, and follows exactly one arrow for that symbol. Each room needs one outgoing R arrow and one outgoing B arrow. It answers YES only if it ends in a starred room. Its room is its entire memory. The target: number of R cards is a multiple of 3, including 0; B cards do not count.

1. Try a two-room design. Mark start and YES rooms and both arrows from every room. Ask a partner for short strings that might break it. Start with the empty string, B, R, RR, RBR and RRR.

Your machine: mark start, YES rooms and R/B arrows



2. A machine passes EVERY string of length at most 2. Its rooms are START (starred) and OTHER (not starred); B stays put, and every R goes to OTHER. Find its shortest failing string. What did the short tests fail to prove?



Shortest failing string / what short tests leave open

Build a certificate about memory

How many rooms must a machine remember?

Follow labeled arrows; remainders; compare histories using common futures.

You may now use three rooms. A history is the part of a string already read. A suffix is the SAME future string attached to each of two histories.

3. Build a correct three-room machine. Explain what each room remembers and prove your explanation survives the next R or B.

Your machine: mark start, YES rooms and R/B arrows

4. Consider the histories empty, R and RR. For EACH pair, find one common suffix that makes exactly one completed string acceptable. Why does that rule out every two-room machine?

Histories	Common suffix	Two answers
empty / R		
empty / RR		
R / RR		

Why every smaller machine fails

Remember two things at once

How many rooms must a machine remember?

Optional: remainder pairs and a systematic lower-bound proof.

New target: the number of R cards is a multiple of 3 AND the number of B cards is even. Zero counts satisfy both requirements. Every finite input length is allowed.

5. Design a machine for this new target. Choose your room layout. Test RRR, BBB, RBRBRBB and the empty string. Explain what a room needs to remember.

Your machine: mark start, YES rooms and R/B arrows

6. Can fewer rooms work? Use the six histories empty, R, RR, B, RB, RRB. Explain how to choose a common suffix that separates ANY chosen pair.

Your construction / explanation

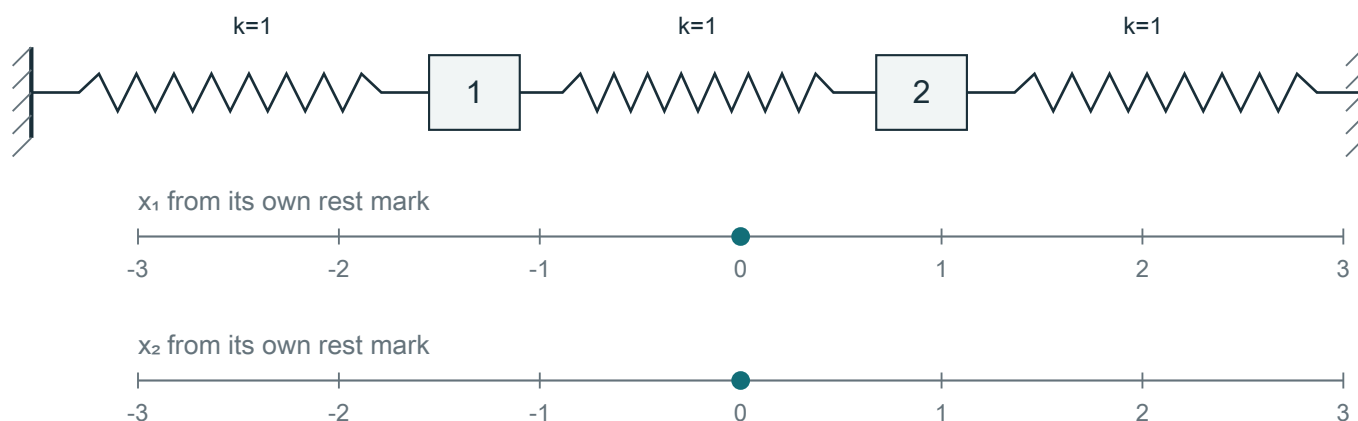
Which shapes pull straight back?

Two motions hiding in one machine

Algebra for force modes; calculus for oscillation and exact nonrepetition.

Two equal unit masses move horizontally between fixed walls. All three springs have stiffness 1. Displacements x_1, x_2 are measured from each mass's own rest mark; positive is right for BOTH. The ideal force law is $F_1 = -2x_1 + x_2$, $F_2 = x_1 - 2x_2$. Ignore friction and assume the linear spring law.

1. Choose three nonzero displacement pairs. Draw each pair and its force pair. Try to find a pair whose force is a **NEGATIVE** multiple of the whole displacement pair. Find a second essentially different shape.



Chosen pairs → force pairs / shared multiplier?

2. Can a third shape work? Give a complete algebraic reason, not just more tested pairs. Exclude the zero pair when discussing a shape.

Your construction / explanation

A different pair of coordinates

Two motions hiding in one machine

Algebra for force modes; calculus for oscillation and exact nonrepetition.

Now use the two shapes you found as building blocks. Add displacement pairs coordinate by coordinate; this is a mathematical decomposition, not two separate physical copies of the masses.

3. Build the displacement $(3,1)$ from one equal-motion piece and one opposite-motion piece. Then do it for a pair (u,v) of your choosing. Why is the decomposition unique?

Equal-motion piece	+	Opposite-motion piece	=	Chosen pair
--------------------	---	-----------------------	---	-------------

How to recover both pieces / why unique

4. Define $S=x_1+x_2$ and $D=x_1-x_2$. Express F_1+F_2 and F_1-F_2 using S,D . What interaction disappeared in these coordinates?

Your construction / explanation

Can periodic pieces fail to repeat together?

Two motions hiding in one machine

Calculus, cosine periods, irrationality; ODE uniqueness may be supplied.

CALCULUS continuation. Unit mass means acceleration equals force. Release from $x(0)=(3,1)$ with both initial velocities 0. A scalar motion $A \cos(\omega t)$ has second derivative $-\omega^2 A \cos(\omega t)$. Use radians.

5. Use your decomposition to propose the two mass positions $x_1(t), x_2(t)$. Verify both force equations and all four initial conditions by differentiation.

Your construction / explanation

6. Each mode is periodic. Can the combined system return EXACTLY to its starting positions at a positive time? Distinguish an exact proof from a plot that looks close.

Your construction / explanation

7. Change only the middle spring stiffness to $k > 0$. Choose k so that the opposite mode completes exactly TWO cycles while the equal mode completes ONE. Show that the combined motion then repeats.

Your construction / explanation

What is shared?

Add a spring: softer or stiffer?

Algebra: force/extension equations, fractions and recursive networks.

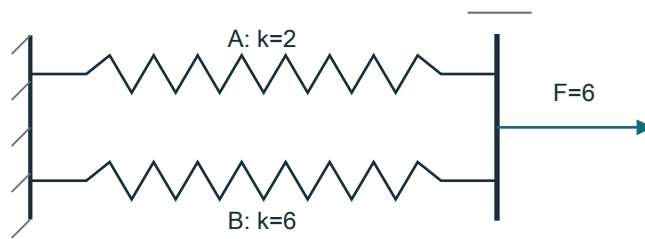
Ideal spring A has $F=2x$; spring B has $F=6x$. Compare END-TO-END with SIDE-BY-SIDE between common rigid guided bars. Connectors are massless, motion is along the springs, and the load is static. Apply total force 6. The bars in the side-by-side drawing translate without rotating.

1. Predict which arrangement extends farther. Draw force and extension labels on EACH spring, then solve. Which quantity must be equal in the two springs for each arrangement?

END-TO-END



SIDE-BY-SIDE



Label forces and extensions / explain what is shared

2. Someone gives each side-by-side spring half the total force. Show exactly what fails. Then explain how adding a spring can make a system either softer or stiffer.

Your construction / explanation

How many three-spring responses exist?

Add a spring: softer or stiffer?

Algebra: force/extension equations, fractions and recursive networks.

Now use EXACTLY THREE identical springs, each $k=1$. A permitted network starts with single springs and repeatedly joins two disjoint smaller networks either in SERIES (end-to-end) or PARALLEL (between common guided bars). Use every spring once; no other connections are allowed. Mirrored drawings count as the same construction type.

3. Build as many different responses as you can. For each, find the total extension under force 6. Compare different-looking drawings that give the same result.

Try networks here; continue on scrap paper. Record each extension.

4. Prove your list of possible responses is complete. Is total extension 6 under force 6 possible with exactly these three springs? If not, give a certificate covering EVERY permitted network.

Your construction / explanation

Turn a design into its reciprocal

Add a spring: softer or stiffer?

Algebra: force/extension equations, fractions and recursive networks.

A dual design swaps EVERY series join for a parallel join, and every parallel join for a series join; each unit spring stays a unit spring. Use the four verified three-spring designs as your first experiments.

5. Pair each three-spring design with its dual. What happens to stiffness and to extension under unit force? Explain from the two combination rules.

Original constructions

Their duals / general rule

6. Use EXACTLY FOUR unit springs to achieve extension 6 under force 6. Find two visibly different arrangements and certify both. Explain why this does not contradict your three-spring impossibility proof.

Arrangement 1 / certificate

Arrangement 2 / certificate

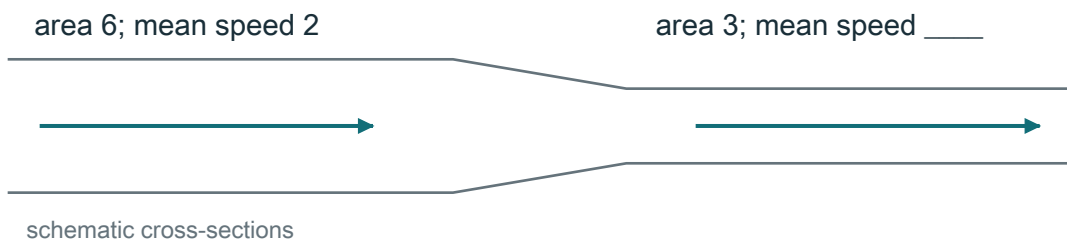
Catch an impossible label

What can a flow map actually tell you?

Core: area, speed and volume per time

A rigid full tube carries constant-density liquid steadily, with no leaks. At a cross-section, $Q = \text{area} \times \text{mean forward normal speed}$. Q is volume per second. If speeds vary, use the area-weighted mean.

1. The wide section has area 6 and mean speed 2. The narrow section has area 3. Invent one possible and one impossible narrow mean-speed label. For the impossible label, explain where the volume count fails after one second.



One second: entering / leaving / discrepancy

2. Split the narrow section into patches of areas 1 and 2, each with a uniform forward speed. Design two different nonnegative speed pairs that carry the required flow. Then test patch speeds 2 and 5. Is the ordinary average of those two readings the section mean?

Narrow-section patches

area 1	area 2
speed ____	speed ____

Two designs / check the 2 and 5 readings

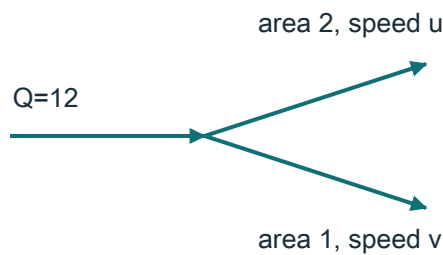
A branch does not choose its own flow

What can a flow map actually tell you?

Core continuation: classify a nonnegative family

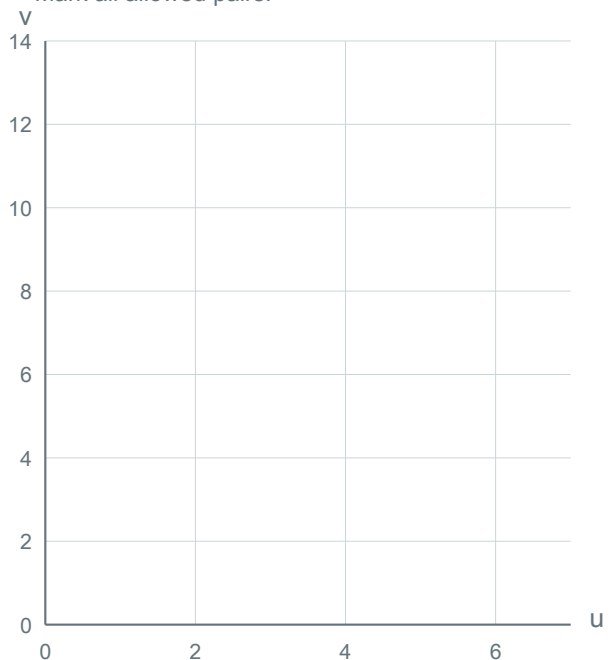
The same incoming volume rate 12 now splits into two steady outlets with areas 2 and 1. Their mean forward speeds are u and v . There are no other outlets or storage changes.

3. Design two conserving splits: in one, the smaller outlet is faster; in the other, it is slower. Describe EVERY nonnegative pair (u,v) that passes the conservation test. What are the largest possible u and v ?



Two splits / equation / endpoints

Mark all allowed pairs.



4. A designer says, “The areas alone determine both speeds.” Refute that claim with a certificate. Then a sensor reports $u=4$. What does this additional observation determine? Does it determine the pressure?

Your construction / certificate

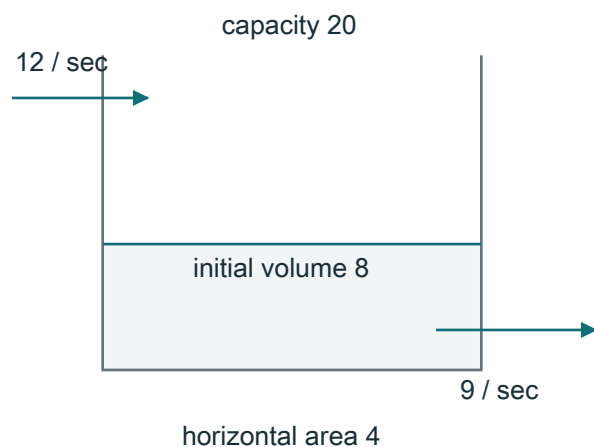
Which assumption changed?

What can a flow map actually tell you?

Continuation: storage and mass density

Keep an observed inlet volume rate 12 and outlet volume rate 9. Change the model explicitly in each task. Rates and dimensions use consistent units.

5. Replace the rigid full tube by a tank with horizontal area 4, initial volume 8 and capacity 20. The two rates remain constant; there is no leak. Predict the water-level rise per second and the first overflow time. Could the same two flow readings alone prove that a pipe has a leak?



Level / first overflow / what else could explain it?

6. Now use a steady compressible gas in a rigid tube, with uniform density at each section. Inlet density is 1; its area is 6 and mean speed is 2. Outlet area is 3 and mean speed is 3. Choose an outlet density that conserves MASS. Explain why conserving volume would give the wrong test here.

INLET

density 1 · area 6 · speed 2

OUTLET

density ____ · area 3 · speed 3

Mass per second / why the volume test changes

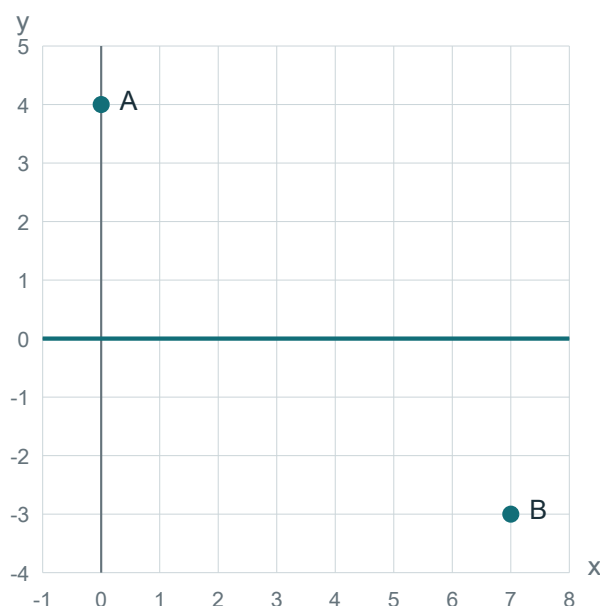
Race time, not distance

Choose the fastest crossing

Calculus core; Pythagoras starts the experiment

$A=(0,4)$, $B=(7,-3)$. A permitted route is a straight segment from A to $C=(x,0)$, then a straight segment to B . Any real x is allowed. Speeds are constant: 3 above $y=0$ and 4 below. Crossing takes no extra time.

1. Choose two crossing points and compare both total distance and total time. Find where the direct straight line from A to B crosses. Must that shortest route also be fastest?



Crossing x / total distance / total time

Crossing x	total distance	total time

2. Build $T(x)$ from the picture. Find a candidate crossing that cannot be improved by a tiny move either way. Give an exact time, not just a graph estimate.

Your construction / certificate

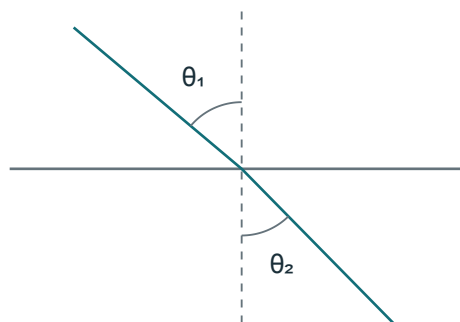
Certify the best route

Choose the fastest crossing

Calculus: a global guarantee and endpoint constraints

A stationary value is a candidate. Your certificate must cover every allowed crossing, including points outside the segment from $x=0$ to $x=7$.

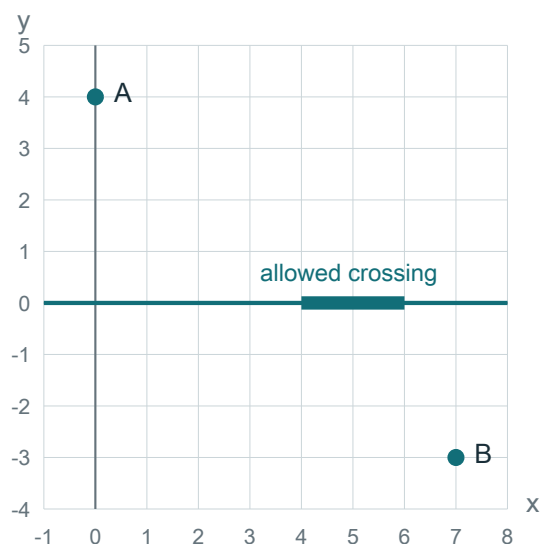
3. Prove that your best crossing is the **ONLY** global minimizer over all real x . Then express its derivative balance using the two angles from the vertical normal.



vertical normal; schematic

Derivative / strict global certificate / angle relation

4. A fence permits crossing only on the closed interval $4 \leq x \leq 6$. Choose the fastest allowed crossing and prove it. Does the derivative have to vanish at this new winner?



Restricted winner / certificate

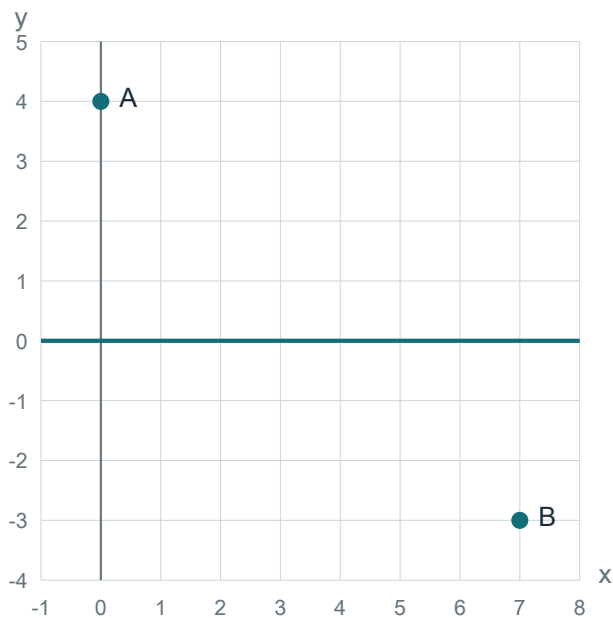
Design the medium to fit your crossing

Choose the fastest crossing

Calculus and algebra: inverse design

Remove the fence. Keep A, B and the upper speed 3. You may choose any finite positive lower speed w .

5. Choose a target crossing s strictly between 0 and 7. Find a lower speed w that makes it the unique fastest crossing. Verify your own choice; also recover the original example.



My target $s = \underline{\hspace{2cm}}$ My lower speed $w = \underline{\hspace{2cm}}$

6. Could a finite positive w make $x=0$ or $x=7$ the unconstrained winner? If w increases from 4, predict which way the winner moves and justify the direction.

Your construction / certificate

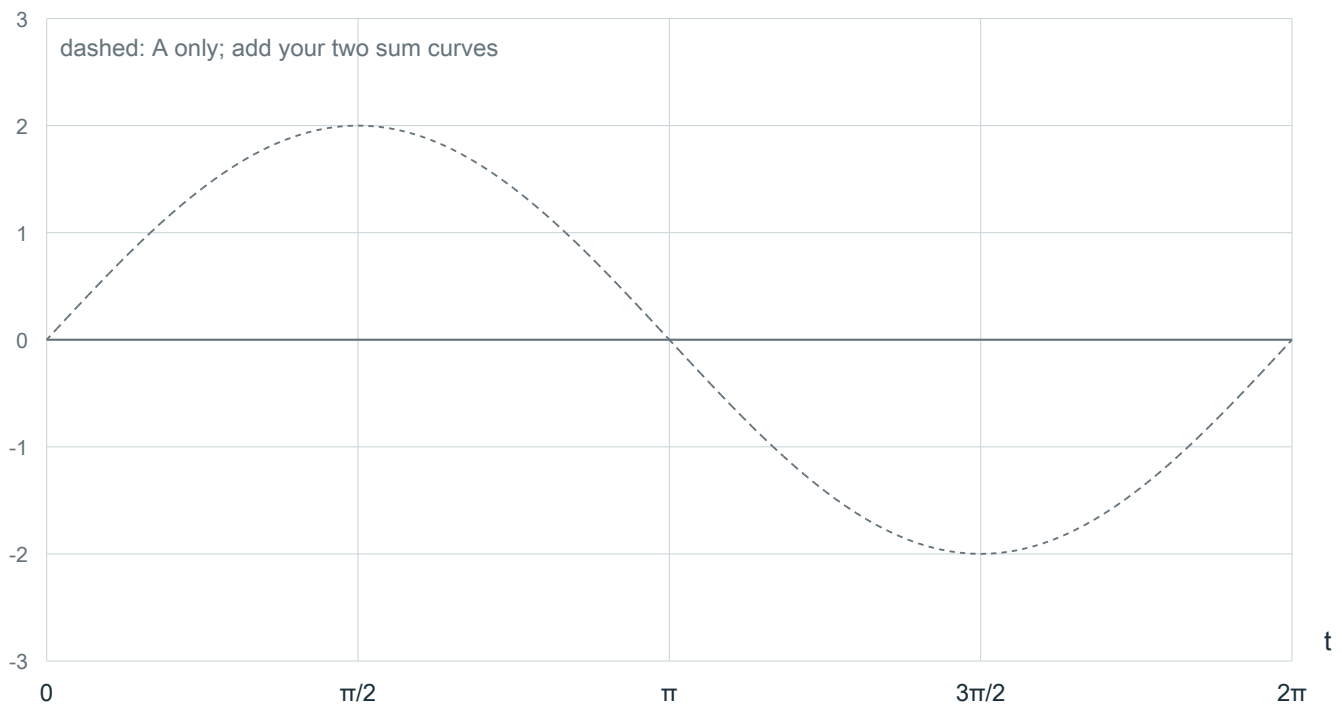
Add a signal and reduce the reading

A dimmer signal, a hidden phase

Core launch: signed heights and squared amplitude

At one detector, two coherent scalar waves have the same fixed frequency and polarization. Their signed readings add. $A(t)=2 \sin t$. You may add $B(t)=\sin(t+\phi)$. Start with $\phi=0$ or π . For $E(t)=C \sin t$, define normalized brightness $I=C^2$.

1. Choose the alignment that you think will dim the detector. Draw the summed waveform over one whole period. Compare A alone with BOTH permitted alignments. Explain using the signed heights, before using brightness.



2. You may now change B's amplitude as well as its phase. Design a nonzero B that makes the total signal zero for the whole period. Does a zero reading here tell you that energy vanished everywhere?

Your construction / certificate

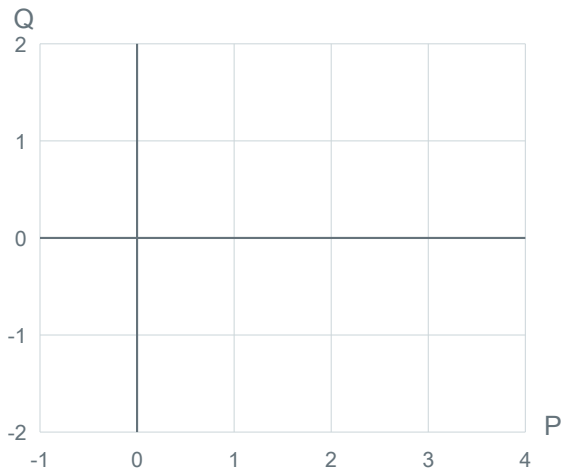
Which brightness values are possible?

A dimmer signal, a hidden phase

Core: sine addition and a complete range

Return to amplitude 1 for B, and allow any real phase ϕ . Define $I=2$ times the average of $E(t)^2$ over a full period. You may use these exact averages: $\text{average}(\sin^2 t)=\text{average}(\cos^2 t)=1/2$; $\text{average}(\sin t \cos t)=0$.

3. Write $E(t)=P \sin t+Q \cos t$. Use the supplied averages to derive I as a function of ϕ . Find the least and greatest possible brightness, and show that every value between them can occur.



Derive I / bound it / attain every value

4. Choose a target brightness strictly between 1 and 9 and find two different phases in $[0, 2\pi)$ that produce it. Describe every phase that makes the detector dimmer than A alone. What information did squaring and averaging lose?

Your construction / certificate

Design a second observation

A dimmer signal, a hidden phase

Core continuation: recover what the first measurement lost

The unknown phase stays fixed while you repeat the experiment on identically prepared signals. You may add a known coherent reference $R(t)$. All intensities use the same full-period normalization and unchanged amplitude scale.

5. A first reading gives $I=5$. Choose ONE reference, $R(t)=\sin t$ or $R(t)=\cos t$, that distinguishes the compatible phases modulo 2π . Predict the second reading in each case. Explain why the other reference fails.

$R(t)=\sin t$

Compatible signals / new readings

$R(t)=\cos t$

Compatible signals / new readings

6. Let J be the second reading after adding $R(t)=\cos t$. Derive a rule that recovers both $\cos \phi$ and $\sin \phi$ from I, J . Decode $I=37/5$, $J=10$. Could $I=5$, $J=6$ come from this exact model? Give a certificate.

Your construction / certificate

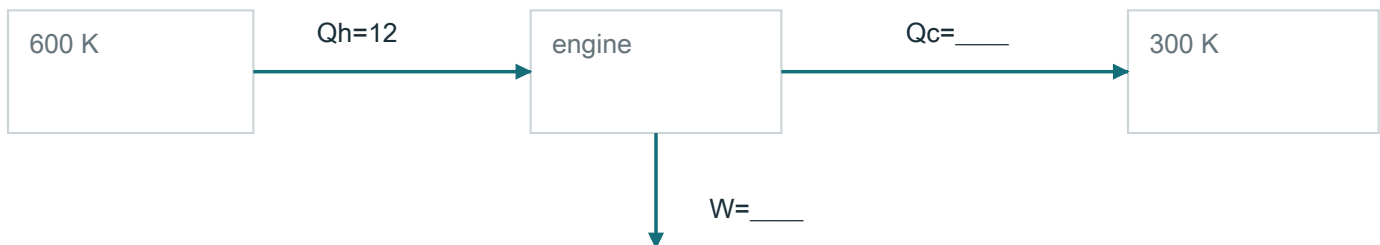
Which claims can survive both tests?

A sales pitch that balances energy and still fails

Core: supplied physical laws and algebra

A cyclic engine takes $Q_h=12$ heat units from a 600 K reservoir, rejects $Q_c \geq 0$ to a 300 K reservoir, and outputs $W \geq 0$. It returns to its starting internal state. Supplied laws: $W+Q_c=12$ and $S_{\text{gen}}=Q_c/300-12/600 \geq 0$. Equality in the entropy law is the reversible ideal case.

1. Invent two energy-balanced claims: one that fails the entropy test and one that passes. Label all three energy transfers and calculate S_{gen} . Then audit the advertised claim $W=8$, $Q_c=4$.



My two proposals / entropy calculations / advertised claim

2. Find the largest work output permitted by both laws, and prove that every larger claim fails. For general $Q_h > 0$ and $T_h > T_c > 0$, derive the corresponding bound. What happens when the reservoir temperatures are equal?

Your construction / certificate

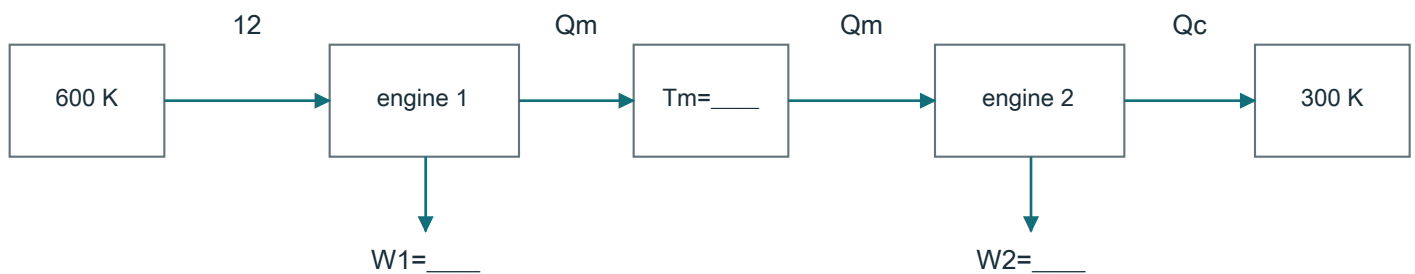
Can a middle reservoir beat the bound?

A sales pitch that balances energy and still fails

Core continuation: reversible cascades

Use two reversible ideal engines. The first works from 600 K down to T_m ; the second works from T_m down to 300 K. All heat rejected by the first is absorbed by the second, so the middle reservoir has no net heat change over the paired cycles. Initial heat remains 12.

3. Choose $T_m=400$ K or 450 K. Compute each engine's work and every transferred heat. Compare with a single reversible engine from 600 K to 300 K.



Transferred heats / work / reservoir changes

4. Prove the total work is independent of T_m for every $300 < T_m < 600$. Could adding more reversible middle stages improve it? Give an argument covering any finite number of stages.

Your construction / certificate

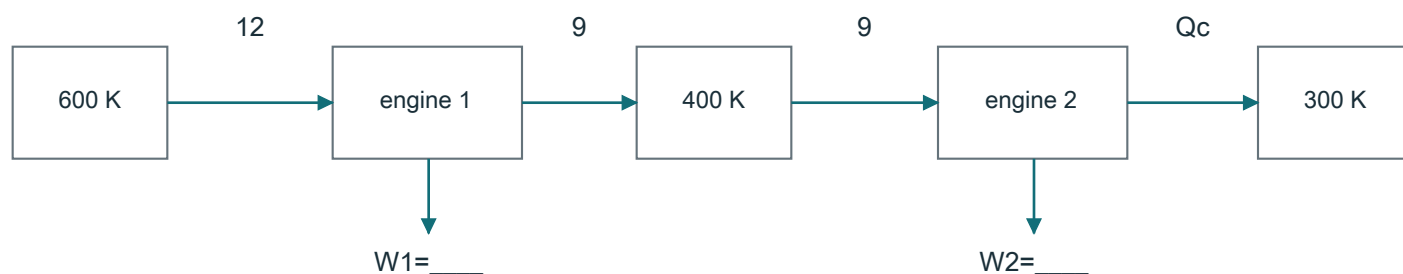
Locate the lost work

A sales pitch that balances energy and still fails

Core continuation: entropy generation as a certificate

Use $T_m=400$ K. A changed first engine takes 12 from 600 K but rejects 9 to 400 K. The second engine is reversible and accepts all 9, then rejects heat to 300 K. Both engines are cyclic; the middle reservoir has no net heat change.

5. Audit each stage and compute total work, final rejected heat and total reservoir entropy change. Compare with the reversible limit. Where did the shortfall arise?



Transferred heats / work / reservoir changes

6. For any cyclic device or cascade exchanging heat only with fixed T_h and T_c , keep the same Q_h and define $S_{gen}=Q_c/T_c-Q_h/T_h$. Derive an exact relation between S_{gen} and the work shortfall from the reversible bound. Can adding a reversible stage erase already positive total entropy generation?

Your construction / certificate

Choose the order, keep the state

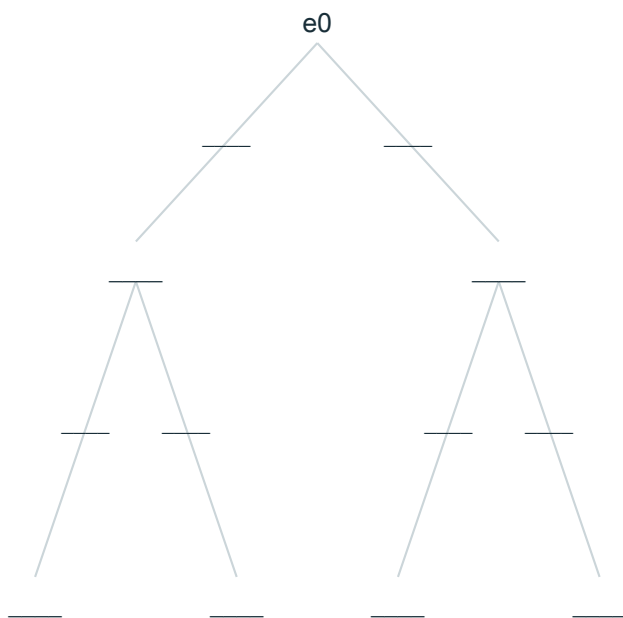
The question between two identical questions

Core: dot products and branch probabilities

Start with the normalized real state $\psi = e_0 = (1, 0)$. Define $e_1 = (0, 1)$. Z measures in $\{e_0, e_1\}$; X measures in $\{+, -\}$, where $\pm = (1, \pm 1)/\sqrt{2}$. Supplied rule: outcome b has probability $(b \cdot \psi)^2$, then the state becomes b . Branch probabilities multiply; disjoint branches add. No evolution occurs between questions.

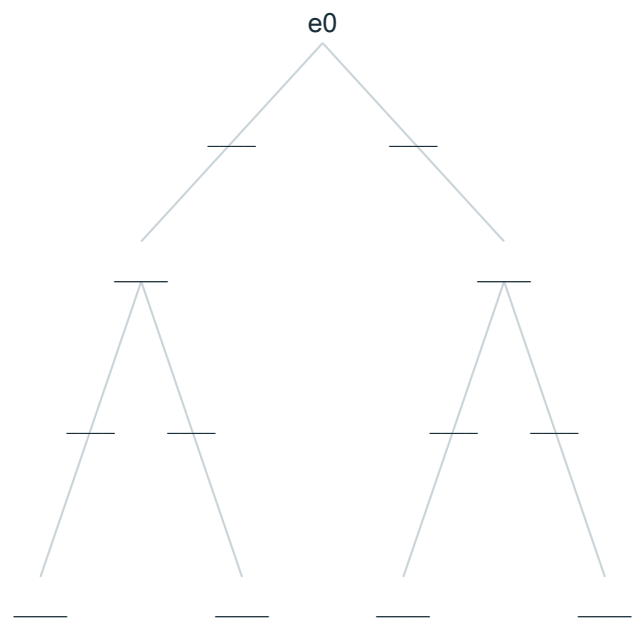
1. Choose X then Z or Z then X . Build its full outcome tree, labeling every branch probability and every new state. Then build the other order. In each experiment, record the Z answer wherever it occurs. How often is that answer e_1 ?

X then Z



Edges: probability. Nodes: new state.

Z then X



Edges: probability. Nodes: new state.

2. Check that probabilities sum to 1 at every live branch. A proposed shortcut computes both questions from the original e_0 and never updates it. Find a specific branch whose probability that shortcut gets wrong.

Your construction / certificate

A certain answer can become uncertain

The question between two identical questions

Core continuation; matrices in prompt 4

Every run again starts in e_0 . Compare a repeated question with a question inserted between repeats. An outcome may be unreported, but the projection update still occurs.

3. Choose $Z \rightarrow Z$ or $Z \rightarrow X \rightarrow Z$. Predict whether the final Z can differ from the first, then compute the exact probability. If the middle X result is erased from the record, does the final Z distribution return to certainty?

$Z \rightarrow Z$ / selected paths and total

$Z \rightarrow X \rightarrow Z$ / selected paths and total

4. Matrix gate: let $P_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $P_+ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} / 2$. Compute $P_0 P_+$ and $P_+ P_0$. Applied to e_0 , the squared norm of each unnormalized vector is the probability of its ordered pair of selected outcomes. Match the two results with your trees.

$P_0 P_+$

apply to $e_0 \rightarrow$ _____

squared norm $\rightarrow$ _____

$P_+ P_0$

apply to $e_0 \rightarrow$ _____

squared norm $\rightarrow$ _____

Match each order to its selected outcomes

Choose the most disruptive single question

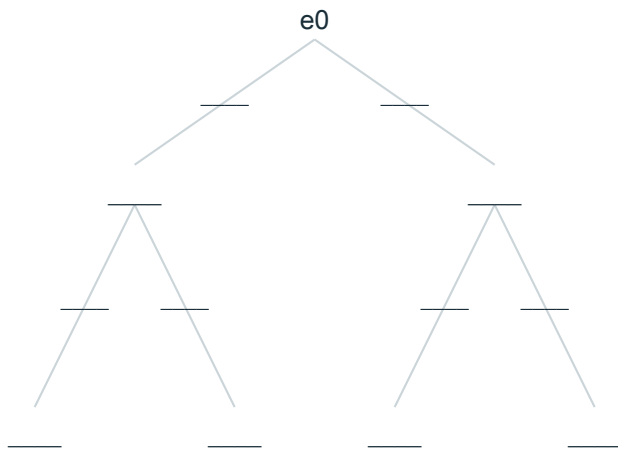
The question between two identical questions

Continuation: trigonometry and a sharp bound

Start in e_0 , ask ONE intermediate question Y, do not retain its outcome, then ask Z. Choose Y's orthonormal basis $b_0 = (\cos \theta, \sin \theta)$, $b_1 = (-\sin \theta, \cos \theta)$, with $0 \leq \theta \leq \pi/2$. The target is final $Z = e_1$.

5. First test the rational basis $b_0 = (3/5, 4/5)$, $b_1 = (-4/5, 3/5)$. Find the target probability using both intermediate branches. Is it larger or smaller than inserting X?

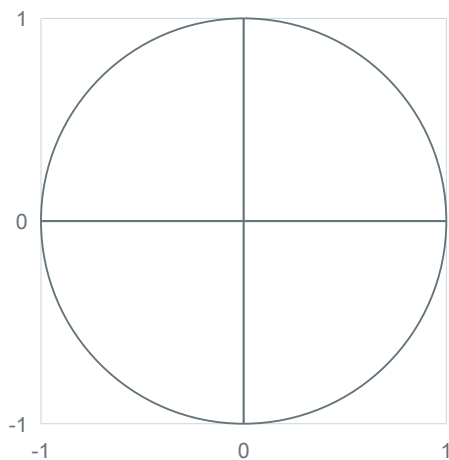
Y then Z



Edges: probability. Nodes: new state.

Both intermediate branches / compare with X

6. Design the angle θ that makes final $Z = e_1$ as likely as possible. Prove that no SINGLE unreported intermediate projective question in this real model can do better. Can the target become certain?



Choose and draw a basis.

Probability for general θ / sharp bound

A favorite arrangement or a favorite class?

Build a sampler for a tiny magnet

Core: four spins and weighted counts

Four labeled sites 1,2,3,4 form an OPEN row, with links 12,23,34 only. Each spin is + or -. If a configuration has a agreeing links, assign weight 2^a . Probability equals its weight divided by the total weight of all configurations.

1. Choose several configurations, including two with the same number of + spins but different neighbor patterns. Find their agreeing links and weights. Does counting + spins alone determine the weight?



a=___ weight=___



a=___ weight=___



a=___ weight=___



a=___ weight=___

Compare equal + counts with different neighbors.

2. Predict which agreement class has the greatest TOTAL weight. Organize all configurations by $a=0,1,2,3$, without counting any twice. Compare the probability of perfect alignment in this model with uniform choice of the sixteen configurations.

a	How many?	Weight each	Total weight
0			
1			
2			
3			

Organized records / alignment probability

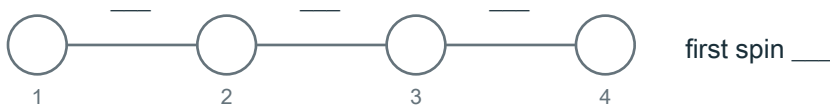
Replace a large raffle by tiny choices

Build a sampler for a tiny magnet

Core continuation: a bijection and an exact probability law

Record a configuration by its first spin and, for each link from left to right, A for agree or D for disagree. The later spins must be reconstructed from that record.

3. Prove that this record determines exactly one configuration and that every configuration has one record. Use it to certify your class counts and total weight without checking sixteen drawings individually.



Reconstruction / why one-to-one / weighted count

4. Design an exact random sampler: choose the first spin fairly, then choose each link independently, with the same chance p of A. What p makes each row's probability proportional to 2^a ? Give a small-ticket implementation and prove the probability of EVERY row.

Fair first spin

choice: ____

Link 12

choice: ____

Link 23

choice: ____

Link 34

choice: ____

My small-ticket design / probability of an arbitrary row

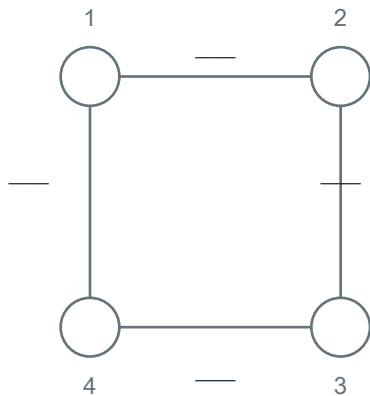
Close the last link

Build a sampler for a tiny magnet

Continuation: parity and conditional sampling

Now add link 41, making a RING of four labeled sites. Weight is still 2^a , but a now counts agreements on all FOUR links. Two configurations are identical exactly when every labeled site has the same spin in both.

5. Try to prescribe the four A/D link symbols freely and reconstruct a spin ring. Which records fail to close? Use that obstruction to find all agreement-class sizes, the total ring weight and the new alignment probability.



Spin values go inside the circles.

Closing rule / class inventory / total and probability

6. Try this repair: choose a fair first spin and FOUR independent links with $P(A)=2/3$. Reject a record if it fails to close; then redraw all choices independently until one closes. Prove that the accepted ring has the desired weighted distribution. Find the acceptance probability.

draw all choices

test closure

accept OR redraw

Proposal probability / acceptance / conditional probability

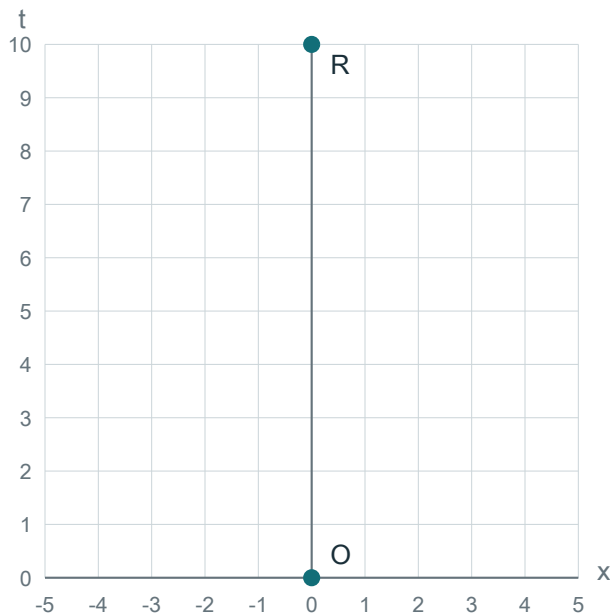
Which turn points are legal?

Choose a route for a clock

Core: a supplied spacetime rule

Events are written (t,x) . The drawing has x horizontal and t vertical. Use units with light speed 1. A massive clock's straight segment must have $\Delta t > 0$ and $|\Delta x| < \Delta t$. Its elapsed proper time is $\sqrt{(\Delta t)^2 - (\Delta x)^2}$. Add proper times over segments. An instantaneous turn is an idealization, not an inertial segment.

1. Departure is $O=(0,0)$, reunion $R=(10,0)$. Choose a turn point $T=(t,d)$ with $d \neq 0$ and draw $O \rightarrow T \rightarrow R$ with both legs legal. Describe the whole region of legal turn points. Test $T=(5,3)$, $(2,3)$ and $(5,5)$.



My route / full legal region / test points

2. Clock A stays at $x=0$. Clock B uses $T=(5,3)$. Compute each clock's total reading at reunion. Then compute the reading for your own legal turn point. Why should measuring the drawn line with a ruler give a different result?

Your construction / certificate

Design a clock reading, then bound every route

Choose a route for a clock

Core continuation: algebra and a global certificate

First restrict to symmetric excursions $O \rightarrow (5, d) \rightarrow R$ with $|d| < 5$. Then allow any finite number of forward-time straight segments, all with speed strictly less than 1, from O to R .

3. Design a symmetric route on which the traveler records exactly 6 time units. Which target readings can symmetric legal routes attain? Can the clock record exactly 0?

Your construction / certificate

4. Could a route with many turns make the traveler's clock read more than A's 10? Give a proof for EVERY permitted finite route, including the equality case.

Your construction / certificate

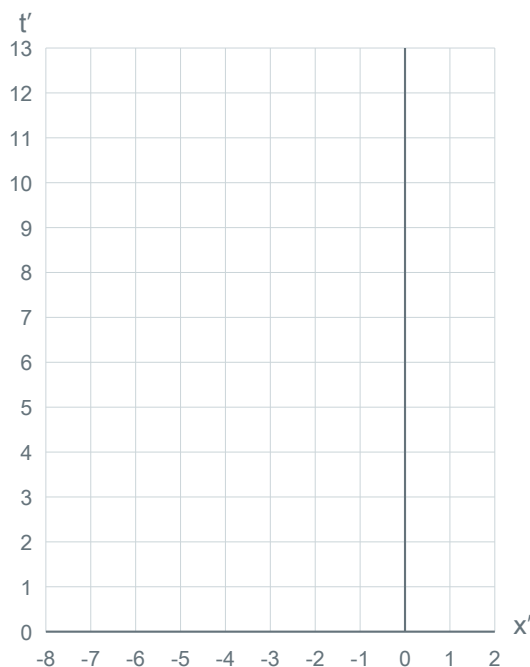
Change the coordinates, keep the events

Choose a route for a clock

Advanced algebra: one inertial boost

Return to $T=(5,3)$. An observer moving at speed $3/5$ uses $t'=(5t-3x)/4$ and $x'=(5x-3t)/4$. This single inertial coordinate system covers the WHOLE trip. Event pairs below are (t',x') ; use the same proper-time rule.

5. Transform O, T and R. Replot BOTH worldlines in the primed grid. Recalculate A's duration and both B-leg durations from the transformed differences. Do the clock readings agree with the first calculation?



Event coordinates / A and both B intervals

6. Expand $(\Delta t')^2 - (\Delta x')^2$ and prove it equals $(\Delta t)^2 - (\Delta x)^2$ for any segment. In the primed frame, which clock changes velocity? Explain why “each sees the other moving” does not make these two complete trips interchangeable.

Your construction / certificate

One center, several witnesses

What can an orbit reveal about an unseen center?

Core: circular-orbit model and algebra

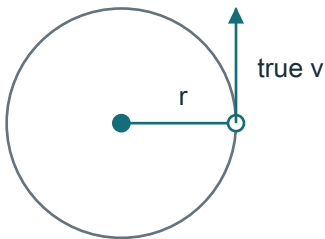
Assume tiny test particles in circular Newtonian orbits about one fixed dominant point mass M . Gravity is the only force. Radii r and speeds v are true three-dimensional values. Supplied circular balance: $v^2/r = GM/r^2$. Use consistent units with $G=1$. All data on this page are exact.

1. Choose any two cards: $(r,v)=(1,6),(4,3),(9,2)$. Infer the central mass from each, then test the third. Explain why the orbiting particle's mass is absent and why a slower outer orbiter need not imply a different center.

true $r=1$ true $v=6$

true $r=4$ true $v=3$

true $r=9$ true $v=2$



one orbit, schematic

Each witness / common center certificate

2. Predict the speed at radius 16. Invent an exact radius/speed card that cannot fit this same M , and give a numerical certificate. Also audit the card $(r,v)=(4,4)$. What if a labeled diameter were mistaken for a radius?

Your construction / certificate

How precise must the next observation be?

What can an orbit reveal about an unseen center?

Core continuation: guaranteed error bounds, not probabilities

For this page radii remain exact, but speeds may lie in stated closed intervals. No distribution or independence is assumed. A common mass is compatible if some speed in each interval satisfies the same circular model.

3. At radius 4 the speed lies in $[2.9, 3.1]$; at radius 9 it lies in $[1.9, 2.1]$. Find EVERY compatible mass. Prove both that masses outside your answer fail and that each mass inside it can fit both observations.

radius 4 observation



radius 9 observation



common masses



Necessity and constructive sufficiency

4. A new experiment must distinguish two exact hypotheses, $M=36$ and $M=64$. You can measure speed at radius 16 with guaranteed absolute error at most ϵ . Find the largest error threshold below which the possible readings are disjoint. What happens at the threshold itself?

$M=36$: possible readings

label center and endpoints

$M=64$: possible readings

label center and endpoints

Disjointness condition / what happens at equality

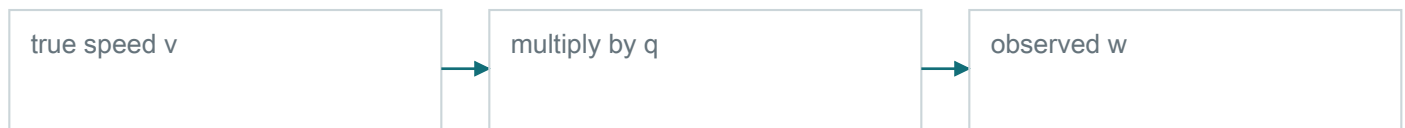
A hidden projection can imitate a different mass

What can an orbit reveal about an unseen center?

Continuation: identifiability and experiment design

Change the observation model explicitly. Radii are still true and exact, but the instrument reports $w=qv$, with the SAME unknown factor $0 < q \leq 1$ for all orbiters. The exact observed (r,w) cards are $(1,6), (4,3), (9,2)$. This common-factor model is stipulated; it is not implied by a drawing.

5. Can these three observations determine M ? Construct two different pairs (M,q) that fit all three. Describe EVERY allowed pair and give a lower bound on M .



Different models / the whole family / lower bound

6. You can also measure the TRUE orbital period T of the radius-4 orbiter, independently of the speed projection. Its circumference is $2\pi r$. Show that one exact period determines M and q . Decode $T=4\pi/3$, using the existing observed $w=3$.

Your construction / certificate

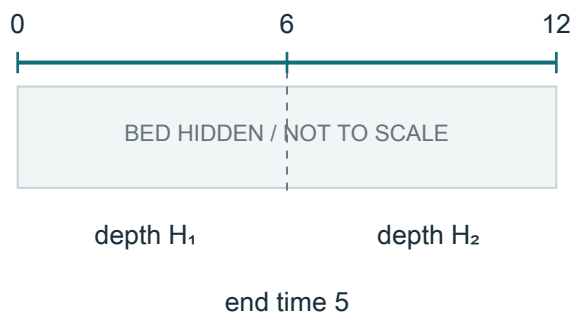
One reading, many hidden channels

Design a survey that can identify the depth

Core: square roots and positive travel times

Use a supplied one-dimensional travel-time model for small-amplitude long waves, no current or rotation. Each region has constant positive finite depth H and speed $\sqrt{gH}$; segment times add. Ignore reflections. Choose units with $g=1$. The channel runs from $x=0$ to 12 ; its depth changes at $x=6$. The end travel time is $T=5$.

1. Choose H_1 from 4, 9 or 16. Find H_2 that gives the measured total time. Compare with a partner who chose differently. Can equal total times certify equal channels?



Chosen depths / verify total time

2. Describe EVERY pair of positive finite depths with $T=5$. Include an equal-depth channel and explain which proposed first depths are impossible.

Your construction / certificate

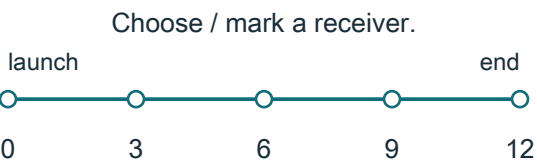
Where should the second receiver go?

Design a survey that can identify the depth

Core continuation: design and invert a measurement

Place one receiver at a chosen x between 0 and 12. It measures S , the first arrival time from the same launch at 0. The end reading remains $T=5$. Let $u=1/\sqrt{H_1}$ and $v=1/\sqrt{H_2}$.

3. Choose $x=3, 6$ or 9 . A test channel gives $S=3/2, 3$ or 4 respectively. Recover both depths from your chosen reading and T . Explain why the new equation supplies information the end reading lacks.



My extra information

Recover / explain new information

4. For an arbitrary receiver position x , decide exactly which positions identify both depths. Give recovery formulas and the restrictions on S for positive finite depths. Include $x=0$ and $x=12$.

Your construction / certificate

A unique answer can still be sensitive

Design a survey that can identify the depth

Continuation: deterministic error bounds

The end time $T=5$ is exact. The inside receiver reports $\hat{S}$ with unknown error $e=\hat{S}-S$ satisfying $|e|\leq 1/20$. The same absolute error bound applies at every position. Recover using $\hat{S}$ as though it were exact. Slowness means reciprocal speed, not depth.

5. Compare placing the receiver at $x=3$ with report $\hat{S}=3/2$, or at $x=6$ with report $\hat{S}=3$. Find every positive depth pair consistent with each report and error bound. Which placement gives the tighter slowness uncertainty?

$x=3$, reported time $3/2$

u _____

v _____

Record endpoints and how u, v are linked.

$x=6$, reported time 3

u _____

v _____

Record endpoints and how u, v are linked.

6. Choose ANY interior x . Derive the largest absolute error in either recovered slowness caused by $|e|\leq\epsilon$, with T still exact. Prove which receiver position minimizes this error amplification. Does a good position remove the wave-model assumptions?

Your construction / certificate

Build a plan worth challenging

Sell products, then price a proof

Core: whole products and resource accounting

Product A uses 2 red and 1 blue counter and sells for 3 points. Product B uses 1 red and 2 blue and sells for 4 points. You have 5 red and 4 blue. Counters cannot be reused. On this page make whole products only.

1. Choose a production plan, trade recipes with a partner, and try to improve the revenue. Record the counters used and left over. Give your best plan and a challenge another plan must beat.

A    → 3 points

B    → 4 points

Plan / ingredients used / leftovers

2. Someone says, “B sells for more, so making as many B products as possible must be best.” Give a counterexample using this stock. What would a proof against ALL competing plans need to control?

Your construction / certificate

Invent prices that certify your plan

Sell products, then price a proof

Core proof: fractions and inequalities

Assign nonnegative audit prices r to one red counter and b to one blue counter. These are proof tools, not extra income. Require the ingredients of each recipe to be worth at least its sale value. For this page fractional quantities of products are also allowed.

3. Design audit prices that make the total stock worth exactly your 10-point plan. Check both recipe inequalities. Prove these prices bound every feasible plan, including fractional ones.

one red: $r =$

one blue: $b =$

Check recipes / prove universal bound

4. Prove whether another fractional plan can also earn 10. Explain which equality conditions in your price proof matter.

Your construction / certificate

When a fractional certificate leaves a gap

Sell products, then price a proof

Continuation: whole-number restrictions

Change the stock to 4 red and 4 blue. Keep the same recipes and sale values. First allow fractional production; then restore the whole-product rule.

5. Find the best fractional plan and certify it. Then find the best whole-product plan and prove that no whole plan does better. Explain why the fractional certificate alone does not finish the second task.

Your construction / certificate

6. You may add ONE red or ONE blue counter to the 4-and-4 stock. Choose which addition gives more whole-product revenue. Certify both resulting optima. Compare the gain with what the audit price of that counter predicts for fractional production.

add RED / plan / upper bound / gain

add BLUE / plan / upper bound / gain

Choose a bag; let a rival challenge it

Build a randomizer your opponent cannot exploit

Core: independent hidden actions and averages

Row chooses U or D; Column chooses L or R. Choices are simultaneous and privately sampled, using independent randomizers. Column pays Row the table entry: U/L pays 2, D/R pays 1, the other entries pay 0. Row wants more; Column wants less. Both know each bag recipe, never the other draw.

1. Build a Row bag with at most six equally likely slips, each marked U or D. Test its EXPECTED payment against a Column player who always chooses L, then one who always chooses R. A rival chooses the worse opponent for your bag. Revise your bag if you can.

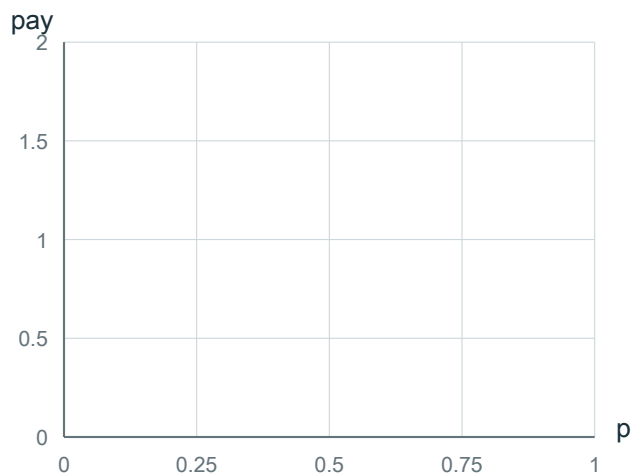
Column pays Row

	L	R
U	2	0
D	0	1

My bag: _____

My bag / expected payments / revision

2. Now allow any real p between 0 and 1 as the probability of U. Find the largest payment Row can guarantee against EVERY independent Column mixture. Prove which p attain it.



Opponent comparisons / optimality proof

The opponent needs a certificate too

Build a randomizer your opponent cannot exploit

Core proof: matching lower and upper bounds

Let q be Column's probability of L, independently of Row's action. A Column guarantee is an upper bound on what Row can earn against that fixed distribution.

3. Design a Column randomizer that keeps EVERY Row mixture at or below the guarantee you found. Prove it works and decide whether its probability q is unique. What do the two certificates establish together?

Design your randomizer

$P(L) = \underline{\hspace{2cm}}$

Upper bound / uniqueness

4. Suppose Column advertises $q=3/4$ and really samples independently from it. Choose Row's best response and compute its payoff. Why can a strategy with a larger payoff here have a worse worst-case guarantee?

Your construction / certificate

How much imperfection can the bag tolerate?

Build a randomizer your opponent cannot exploit

Continuation: inequalities and information rules

Keep independent private sampling for the first task. Then change what the opponent can see, and state that change explicitly.

5. Choose $0 \leq \epsilon \leq 1/3$. Find EVERY p that guarantees Row at least $2/3 - \epsilon$, and EVERY q that keeps Row at most $2/3 + \epsilon$. Test $\epsilon = 1/15$. Prove both necessity and sufficiency.

Row p



Column q



Endpoints / necessity / sufficiency

6. Change the rules: Column sees Row's actual action before choosing. Design Column's response and determine Row's best guarantee. Then suppose both instead use the SAME uniform ticket from $\{1, 2, 3\}$: Row uses U only on 1 and Column uses L only on 1. Compute the payoff and explain why the independent-mixture formula cannot be used.

Your construction / certificate

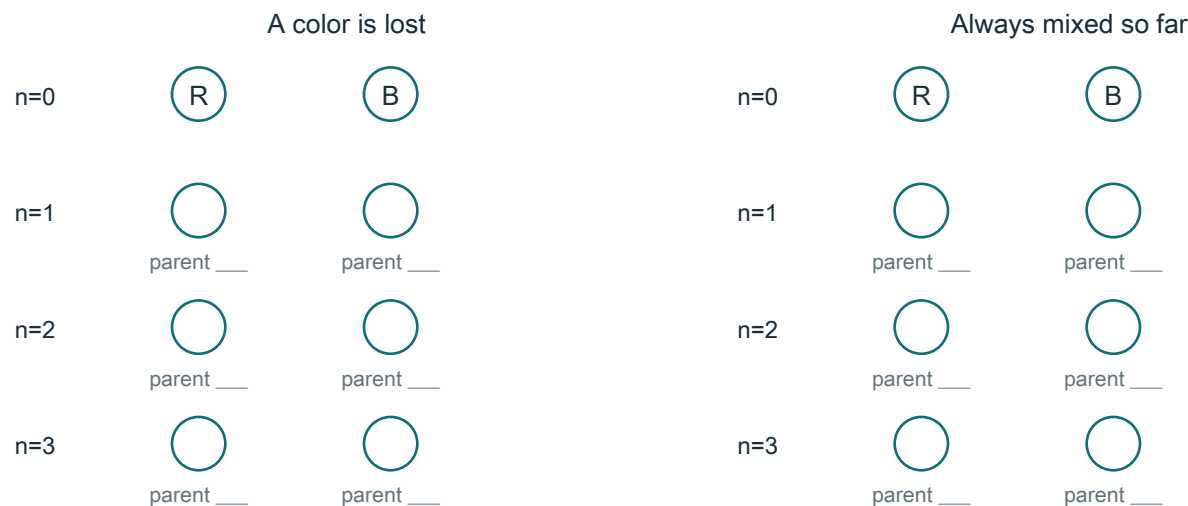
Build two possible family histories

A fair copying rule can erase a color

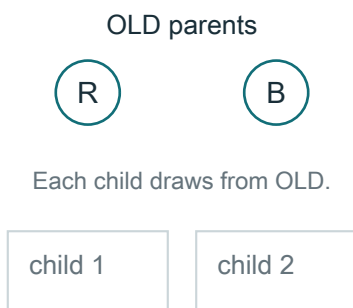
Core: copy from the OLD generation

Start with two parents: one red and one blue. To make each of two children, independently choose one parent uniformly, with replacement, and copy its color. Both children choose from the unchanged old generation. Only then replace the parents by the two children. Use fresh independent draws each generation. No mutation or color advantage occurs.

1. Choose and draw a legal three-generation history that loses a color, and another that stays mixed throughout. Name the selected parent for each child. Does either history contradict the fair copying rule?



2. From one red and one blue, list the ordered parent choices for one generation and their probabilities. Group them by the new red count. Find the expected red count. Then give the transition rule from an all-red or all-blue population.



Ordered choices / probabilities / mean

A mean is not a typical history

A fair copying rule can erase a color

Core proof: finite probabilities and a limit

Let X_k be the red count after k generations from the original mixed pair, with $X_0=1$. Once a color is lost, it cannot return under this rule.

3. Find exact probabilities for $X_k=0, 1$ and 2 for EVERY integer $k \geq 0$. Prove that the probability of remaining mixed forever is zero. Is there a finite generation by which loss is absolutely certain?

Your construction / certificate

4. Choose k so that the chance of still being mixed is below $1/100$. Compute the expected red count at that k and explain how it can coexist with almost every population having just one color. What changes if the two parent draws are WITHOUT replacement?

Your construction / certificate

What if the colors start unequally?

A fair copying rule can erase a color

Continuation: three independent offspring

Now use THREE parents and THREE children, with the same independent with-replacement rule. Start with one red and two blue. Write X_k for the red count, so $X_0=1$. Counts 0 and 3 are absorbing.

5. Find the transition probabilities from red count 1 and from red count 2. Prove that the next expected red count equals the current count. Use your rows to find the chance of remaining mixed for k generations.

Fill each row; check its sum.

from	to 0	to 1	to 2	to 3
1				
2				

6. Prove the eventual all-red probability from the one-red start. Do not argue by red/blue symmetry: the starting population is unequal. Explain why this is a probability across histories, not a fractional final population.

Your construction / certificate

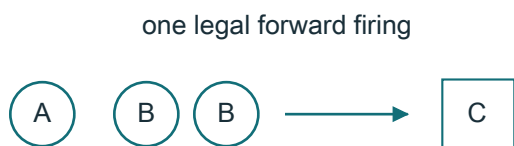
The counter total changes. What survives?

Design labels that survive a reaction

Core: reversible legal trades

A forward move consumes 1 A and 2 B and creates 1 C. A backward move consumes 1 C and creates 1 A and 2 B. A move is legal only when all consumed counters are present. Start with $(A,B,C)=(4,7,1)$. Counts must stay nonnegative integers.

1. Explore both directions. Choose a target you can reach and record a legal sequence. Compare the total number of counters before and after a forward move. Could total counter count certify that a target is impossible?



Only fire when ingredients exist.

My target / legal firings / count change

2. Put a positive whole-number weight on each species. Design TWO different weight systems for which every legal move preserves total weight. Make the two systems more than just scaled copies. Test them on your sequence.

Weight system I

A: ____ B: ____ C: ____

Total before / after:

Weight system II

A: ____ B: ____ C: ____

Total before / after:

Turn a label into a complete certificate

Design labels that survive a reaction

Core proof: arbitrary weights and impossible targets

Allow weights to be any real numbers, including zero or negative values. A label total is $w_A A + w_B B + w_C C$. It must be preserved by the reaction wherever that reaction is legal, not just at one chosen inventory.

3. Find EVERY weight triple that is always conserved. Find two simple conserved totals from which every other linear conserved total can be built. Prove that your list is complete.

Your construction / certificate

4. Find EVERY reachable state from $(4,7,1)$, and draw legal arrows connecting your list. Prove completeness as well as reachability. Invent one nonnegative target that passes the $A+C$ test but fails $B+2C$, and another that does the opposite.

Arrange your states and draw legal arrows; use more space if needed.

When do invariant tests finish the job?

Design labels that survive a reaction

Continuation: arbitrary inventories and a changed reaction

First keep $A+2B \leftrightarrow C$ but allow any initial nonnegative integer inventory (a,b,c) . Then change the reaction explicitly in the second task.

5. Classify EVERY reachable state from (a,b,c) . Give a legal route to each state you claim and the smallest possible number of reaction firings needed to reach it.

Your construction / certificate

6. Replace the reaction by $2A \leftrightarrow 2B$. Starting at $(A,B)=(5,0)$, find all reachable states. Why does the conserved total $A+B$ fail to decide reachability by itself? State an extra invariant and prove that your full list is reachable.

Your construction / certificate

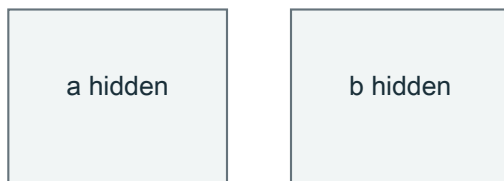
Different tanks, the same entire record

Choose when one sensor can separate two tanks

Core: hidden states and a fixed output law

At time $n=0$ the nonnegative real amounts are a and b . Each step multiplies BOTH amounts by $1/2$. The sensor reports only their sum y_n . This geometric decay rule is supplied; it is not a law for every physical tank.

1. Choose two different nonnegative starting splits with total 8. Predict their entire sensor records and prove whether any later reading can distinguish them. Describe every initial state giving this record. Is the zero-total record exceptional?



No drawn fill levels are data.

sensor: sum only

Two hidden starts / entire record

2. Keep the equal decay rates. You may take one extra reading at time 0: either a alone or $z=a+2b$. Choose one sensor and prove that it recovers every initial pair when combined with the total y_0 . Test your chosen split.

Your construction / certificate

Different rates can reveal the hidden split

Choose when one sensor can separate two tanks

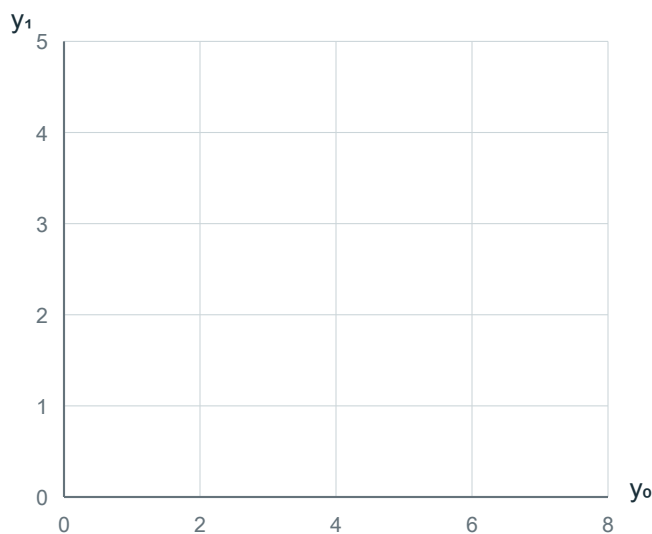
Core proof: reconstruction and feasible readings

Change the known rates: the first amount retains $1/2$ each step, the second retains $1/4$. The sensor still reports only $y_n = a(1/2)^n + b(1/4)^n$. Initial amounts remain nonnegative real numbers.

3. The readings are $y_0=8$ and $y_1=3$. Recover both starting amounts and predict y_2 . A rival proposes (6,2). Use an actual sensor reading to reject or accept that proposal.

Your construction / certificate

4. Derive a reconstruction formula for arbitrary readings y_0, y_1 . Decide exactly which pairs of readings come from nonnegative amounts. Include zero amounts and prove sufficiency.



All feasible readings / include boundaries

Choose rates and timing with noise in mind

Choose when one sensor can separate two tanks

Continuation: exact deterministic error bounds

The initial total y_0 is exact. One later reading has additive error e with $|e| \leq 1/100$. This absolute error bound is the same at every observation time. All rates are known exactly; compute the inverse from the reported reading.

5. Keep the first rate $1/2$. Choose the second rate as $1/4$ or $49/100$. For a true initial state $(4,4)$, predict the time-1 reading and find the greatest possible error in either reconstructed amount. Prove a formula explaining the difference.

Your construction / certificate

6. Now choose when to observe. Rates are $r=3/4$ and $s=1/2$. Use y_0 and one reading y_k , where k is any positive integer. Find the k that gives the smallest worst-case reconstruction-error bound, and prove it beats EVERY other allowed k .

Choose trial delays; then explain what happens for all later delays.

chosen k	reconstruction / evidence	error bound
------------	---------------------------	-------------

Invent a code that an adversary cannot confuse

Four messages through a damaged strip

Core: choose patterns and try to break them

A message is sent as a fixed-length strip of 0s and 1s. An adversary may flip at most ONE bit, with its position hidden, or make no change. You need four distinct messages. The receiver knows the codebook, but not which message was sent.

1. Choose four five-bit codewords for your messages. Let a partner try to produce the SAME received strip from two different messages using at most one flip from each. Revise your code if that happens. Record a successful code or the strongest obstruction you found.

A

--	--	--	--	--

B

--	--	--	--	--

C

--	--	--	--	--

D

--	--	--	--	--

Use the rest for revisions or an ambiguous received strip.

2. How far apart must EVERY pair of codewords be to make one-bit correction possible? Prove necessity by constructing an ambiguous received word when the condition fails, and prove sufficiency for every allowed error.

Your construction / certificate

Audit a construction, then rule out a shorter one

Four messages through a damaged strip

Core proof: correction and a counting obstruction

Now compare with this supplied codebook: A=00000, B=11100, C=10011, D=01111. It is a construction to verify, whether or not it matches your first design. Keep the at-most-one-flip promise.

3. Audit all six pairs. Decode 10111, 01000 and 11100, explaining every correction or no-change decision. Then design a short universal decoding instruction for this codebook.

A	0	0	0	0	0
B	1	1	1	0	0
C	1	0	0	1	1
D	0	1	1	1	1

Keep the at-most-one-flip promise.

Pair distances / decode / rule

4. Could ANY four-bit codebook carry four messages with the same guarantee? Count the received words possible from one codeword and use your disjointness proof. Explain why this also rules out every shorter length.

Your construction / certificate

What can the receiver honestly conclude?

Four messages through a damaged strip

Continuation: promises, detection and correction

Use the supplied five-bit codebook. A decoder is allowed to say “the error promise failed” when no codeword lies within one flip.

5. How many five-bit received words lie within one flip of this codebook? Find a received word outside all those neighborhoods and prove that it violates the promise. Then show how TWO flips can fool the decoder into returning the wrong message.

An outside word / evidence / a two-flip deception

--	--	--	--	--

outside

--	--	--	--	--

sent

--	--	--	--	--

received

6. Change the goal: detect any single flip, but do not necessarily correct it. Construct four messages using THREE bits and prove that every one-bit change is detected. Exhibit an ambiguity showing why detection is weaker than correction.

A

--	--	--

B

--	--	--

C

--	--	--

D

--	--	--

Detection proof / correction ambiguity

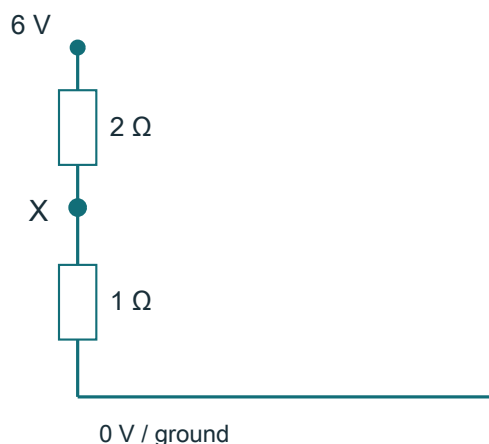
The meter is another branch

Design a meter that disturbs the circuit less

Core: test a voltage using current balance

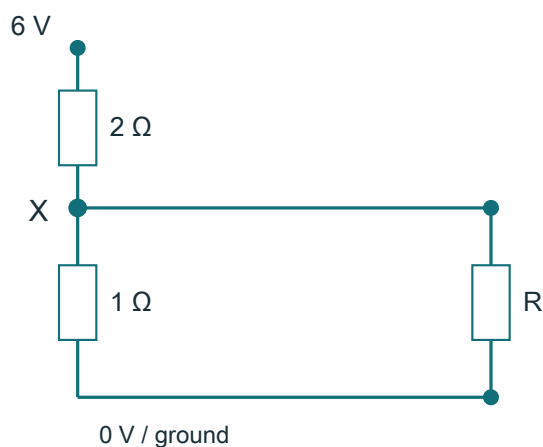
An ideal source fixes the top node at 6 V relative to ground at 0 V. A $2\ \Omega$ resistor joins the top node to X; a $1\ \Omega$ resistor joins X to ground. Supplied laws: current through resistance R is voltage drop/R; incoming current equals outgoing current at X. A meter of resistance $R > 0$ joins X to ground and reports voltage X.

1. With NO meter attached, choose a trial voltage between 0 and 6 and compute both currents. Adjust until current balances. Find the exact voltage X and explain why it is unique.



Trial voltage / balance / uniqueness

2. Choose meter resistance $R=1, 2$ or $6\ \Omega$. Predict the direction of the change, then solve the LOADED node equation. Find all three currents and compare their sum at X with the no-meter result.



My R / voltage / three currents

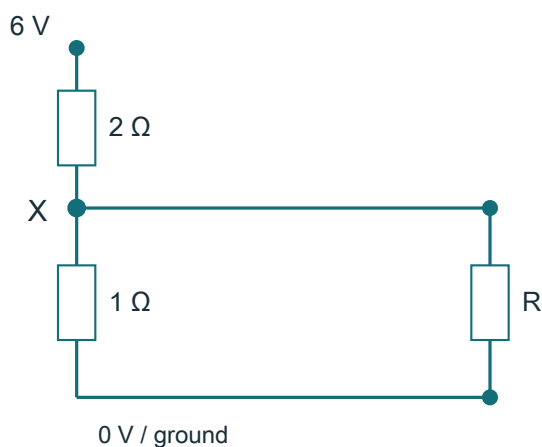
Specify how accurate the meter must be

Design a meter that disturbs the circuit less

Core proof: a sharp design inequality

Allow any real meter resistance $R > 0$. Define relative loading error as $E = (2 - X)/2$, comparing the loaded reading with the exact no-meter voltage 2 V. Component values and the source remain exact.

3. Find X and each branch current as formulas in R . Prove whether ANY finite positive R gives exactly the no-meter voltage. Explain the ideal limit as R becomes arbitrarily large.



Formulas / finite R / ideal limit

4. Choose a tolerance $0 < \epsilon < 1$. Find the smallest R that guarantees $E \leq \epsilon$, proving both directions. Test a 1% tolerance. Does meeting this design bound account for every possible real-world measurement error?

Your construction / certificate

Agreement is not the same as non-disturbance

Design a meter that disturbs the circuit less

Continuation: energy and several meters

Supplied power laws: the source supplies $6 I_{in}$ watts, and a resistor dissipates $I^2 R$, equivalently voltage-drop²/ R . Multiple meters are independent branches electrically, not independent copies of an undisturbed quantity.

5. For $R=2\ \Omega$, audit source power against the sum of all three resistor dissipations. Then model $R=0$ as an ideal wire from X to ground, separately from the positive-R formula. Find its node voltage, currents and resistor-power balance.

Power audit / then treat the ideal wire separately

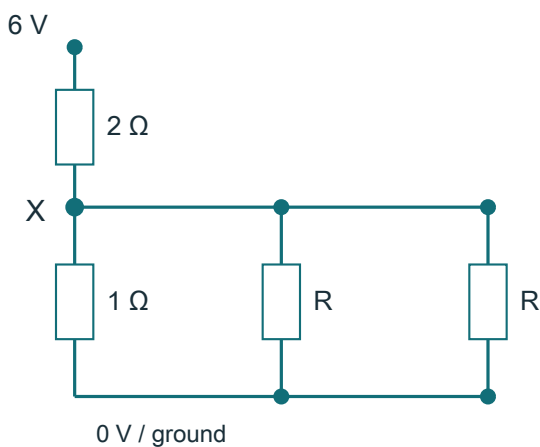
source

$2\ \Omega$

$1\ \Omega$

meter

6. Attach TWO meters, each $R=2\ \Omega$, across X and ground. They agree with each other: do they reproduce the no-meter voltage? Generalize to m identical meters of resistance $R>0$, where m is a positive integer, and design R so the combined loading error is at most ϵ .



Two readings / m branches / design bound